

$$J(x) = \int_0^1 \left(\frac{1}{2} \dot{x}(t)^2 + 5x(t)\dot{x}(t) + x(t)^2 + 5x(t) \right) dt$$

$$\text{find } x^*: [0, 1] \rightarrow \mathbb{R} \quad \begin{aligned} x^*(0) &= 1 \\ x^*(1) &= 3 \end{aligned}$$

$$\text{Set } u = \dot{x}$$

$$\therefore J(x, u, t) = \int_0^1 \left(\frac{1}{2} u^2 + 5xu + x^2 + 5x \right) dt$$

$$\text{want: } \min_u J(x) \quad \text{s.t.} \quad \begin{aligned} \dot{x} &= u \\ x(0) &= 1 \\ x(1) &= 3 \end{aligned}$$

$$\therefore f(x, u, t) = \dot{x} = u$$

$$h(x(t_f), t_f) = 0$$

$$g(x, u, t) = \frac{1}{2} u^2 + 5xu + x^2 + 5x$$

$$\therefore H = g + pf$$

$$\therefore H(x, u, p) = \frac{1}{2} u^2 + 5xu + x^2 + 5x + pu$$

Based on PMP. The NOC is

$$\dot{x}^*(t) = \frac{\partial H}{\partial p} = u^*$$

$$\dot{p}^*(t) = -\frac{\partial H}{\partial x} = -(5u^* + 2x^* + 5)$$

$$u^*(t) = \underset{u}{\operatorname{argmin}} H$$

$$\therefore \frac{\partial H}{\partial u} = 0$$

$$\therefore u^* + 5x^* + p^* = 0$$

$$\therefore u^* = -5x^* - p^*$$

$$\therefore \dot{u}^* = -5\dot{x}^* - \dot{p}^*$$

$$\begin{aligned}\therefore \dot{u}^* &= -5u^* + 5u^* + 2x^* + 5 \\ &= 2x^* + 5\end{aligned}$$

$$\therefore \ddot{x}^* = 2x^* + 5$$

So we need to find a solution of

$$\begin{cases} x''(t) = 2x(t) + 5 \\ x(0) = 1 \\ x(1) = 3 \end{cases}$$

assume $X(t) = C$

$$\therefore X'(t) = 0$$

$$X''(t) = 0$$

$$\therefore 0 = 2C + 5$$

$$\therefore C = -\frac{5}{2}$$

assume $X(t) = Y(t) - \frac{5}{2}$

$$\therefore X'(t) = Y'(t)$$

$$X''(t) = Y''(t)$$

$$\therefore Y''(t) = 2 \left(Y(t) - \frac{5}{2} \right) + 5$$

$$Y''(t) = 2Y(t)$$

$\therefore Y(t)$ must be like e^{rt} .

which r is a number

$$\therefore (e^{rt})' = r e^{rt}$$

$$(e^{rt})'' = r \cdot r e^{rt} = r^2 e^{rt}$$

assume $Y(t) = e^{rt}$

$$\therefore r^2 e^{rt} = 2 e^{rt}$$

$$\therefore r = \pm \sqrt{2}$$

$$\therefore Y_1(t) = e^{\sqrt{2}t} \quad Y_2(t) = e^{-\sqrt{2}t}$$

$$\therefore Y(t) = C_1 e^{\sqrt{2}t} + C_2 e^{-\sqrt{2}t}$$

$$\therefore X(t) = C_1 e^{\sqrt{2}t} + C_2 e^{-\sqrt{2}t} - \frac{5}{2}$$

$$\because x(0) = 1 \quad , \quad x(1) = 3$$

$$\therefore \begin{cases} C_1 + C_2 - \frac{5}{2} = 1 \\ C_1 e^{\sqrt{2}} + C_2 e^{-\sqrt{2}} - \frac{5}{2} = 3 \end{cases}$$

$$\therefore C_2 = \frac{7}{2} - C_1$$

$$\therefore C_1 e^{\sqrt{2}} + \left(\frac{7}{2} - C_1\right) e^{-\sqrt{2}} - \frac{5}{2} = 3$$

$$(e^{\sqrt{2}} - e^{-\sqrt{2}}) C_1 = \frac{11 - 7e^{-\sqrt{2}}}{2}$$

$$C_1 = \frac{11 - 7e^{-\sqrt{2}}}{2(e^{\sqrt{2}} - e^{-\sqrt{2}})}$$

$$\therefore C_2 = \frac{7e^{\sqrt{2}} - 11}{2(e^{\sqrt{2}} - e^{-\sqrt{2}})}$$

$$\therefore x^*(t) = \frac{11 - 7e^{-\sqrt{2}}}{2(e^{\sqrt{2}} - e^{-\sqrt{2}})} e^{\sqrt{2}t}$$

$$+ \frac{7e^{\sqrt{2}} - 11}{2(e^{\sqrt{2}} - e^{-\sqrt{2}})} e^{-\sqrt{2}t} - \frac{5}{2}$$